

Namit Rana

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Achievements

- **Honorable Mention — Google × Kaggle Tunix Hackathon:** Ranked in the **top 0.14% of 11,173 global entrants** by developing an AI reasoning framework using **Gemma 2B**.
- **1st Place Winner — IEEE Vihaan 8.0 Hackathon (DTU):** Outperformed **400+ engineering teams** by engineering an accessibility-first educational platform under intense time constraints.
- **3rd Place Winner — Hack with Gujarat:** Designed and built *MindBoard*, a hands-free device running **Computer Vision** tracking models to enable virtual mouse control for motor-impaired users.
- **Best Paper Award — ICIDSSD'26 (Jamia Hamdard):** Authored peer-reviewed machine learning research optimizing deep learning **CNN** architectures for high-accuracy skin lesion classification.

Projects

UberMicro – Distributed Ride-Sharing Platform | *Node.js, Express.js, MongoDB, RabbitMQ, CloudAMQP*

- Decomposed monolith into **3 independently deployable microservices** behind an API Gateway; replaced synchronous dispatch with **RabbitMQ pub-sub**, removing the blocking service dependency under load.
- Chose **long-polling over WebSockets** for ride status updates, avoiding persistent connection overhead while delivering real-time push on stateless REST services.

FinLedger – Core Banking Backend Engine | *Node.js, Express.js, MongoDB Atlas, JWT, OAuth2*

- Replaced mutable balance fields with an **immutable ledger model** — balance derived via aggregation pipelines from transaction history, ensuring auditability and eliminating update anomalies.
- Enforced **idempotency keys on all transaction endpoints** to prevent duplicate processing on retries; secured API with **JWT + token blacklisting** for stateless logout and deployed to Render.

DevMind – AI-Powered Collaborative IDE | *React, Node.js, Socket.io, Redis, MongoDB, Gemini API*

- Paired **Socket.io with Redis pub-sub** for real-time messaging and presence sync across collaborative rooms, decoupling socket state from individual server instances.
- Injected **Gemini API** with prompt-engineered file-tree context for project-aware AI suggestions; integrated **Web Container API** to run code directly in the browser, eliminating server-side runtime infrastructure.

Technical Skills

- **Languages:** C++, JavaScript, TypeScript
- **Backend & Systems:** Node.js, Express.js, socket.io, Microservices, RabbitMQ, Docker, GitHub Actions (CI/CD)
- **Databases & Infrastructure:** MongoDB Atlas, Redis, AWS (EC2, S3)
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS

Education

University of Delhi

B.Sc. Computer Science

Aug 2023 – May 2026

New Delhi, India

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

Experience

Contract Software Engineer – E-Commerce Platform

Freelance

June 2025 – Present

Delhi, India

- Architected a **full-stack order management system** using Node.js, Express.js, and MongoDB for a textile retailer, designing compound-indexed schemas for efficient high-SKU catalog queries and automating end-to-end order workflows.
- Integrated **WhatsApp Business API** as a real-time lead capture layer, replacing manual inquiry handling; exposed **RESTful APIs** with input validation and role-based access, deployed with environment-isolated config.